

A funcoïd related to directed topological spaces

OPG-60019

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture Let R be the complete funcoïd corresponding to the usual topology on extended real line $[-\infty, +\infty] = \mathbb{R} \cup \{-\infty, +\infty\}$. Let $\geq$ be the order on this set. Then $R \sqcap^{\text{FCD}} \geq$ is a complete funcoïd.

Proposition It is easy to prove that $\langle R \sqcap^{\text{FCD}} \geq \rangle \{x\}$ is the infinitely small right neighborhood filter of point $x \in [-\infty, +\infty]$.

If proved true, the conjecture then can be generalized to a wider class of posets.